



Original Investigation | Otolaryngology

Deep Learning Model of Primary Tumor and Metastatic Cervical Lymph Nodes From CT for Outcome Predictions in Oropharyngeal Cancer

Bolin Song, PhD; Amaury Leroy, PhD; Kailin Yang, MD, PhD; Sirvan Khalighi, PhD; Krupal Pandav, MD; Tanmoy Dam, PhD; Jonathan Lee, MD; Sarah Stock, MD; Xiao T. Li, MD; Jay Sonuga, MD; Pingfu Fu, PhD; Shlomo Koyfman, MD; Nabil F. Saba, MD; Mihir R. Patel, MD; Anant Madabhushi, PhD

Abstract

IMPORTANCE Primary tumor (PT) and metastatic cervical lymph node (LN) characteristics are highly associated with oropharyngeal squamous cell carcinoma (OPSCC) prognosis. Currently, there is a lack of studies to combine imaging characteristics of both regions for predictions of p16+ OPSCC outcomes.

OBJECTIVES To develop and validate a computed tomography (CT)-based deep learning classifier that integrates PT and LN features to predict outcomes in p16+ OPSCC and to identify patients with stage I disease who may derive added benefit associated with chemotherapy.

DESIGN, SETTING, AND PARTICIPANTS In this retrospective prognostic study, radiographic CT scans were analyzed of 811 patients with p16+ OPSCC treated with definitive radiotherapy or chemoradiotherapy from 3 independent cohorts. One cohort from the Cancer Imaging Archive (1998-2013) was used for model development and validation and the 2 remaining cohorts (2002-2015) were used to externally test the model performance. The Swin Transformer architecture was applied to fuse the features from both PT and LN into a multiregion imaging risk score (SwinScore) to predict survival outcomes across and within subpopulations at various stages. Data analysis was performed between February and July 2024.

EXPOSURES Definitive radiotherapy or chemoradiotherapy treatment for patients with p16+ OPSCC.

MAIN OUTCOMES AND MEASURES Hazard ratios (HRs), log-rank tests, concordance index (C index), and net benefit were used to evaluate the associations between multiregion imaging risk score and disease-free survival (DFS), overall survival (OS), and locoregional failure (LRF). Interaction tests were conducted to assess whether the association of chemotherapy with outcome significantly differs across dichotomized multiregion imaging risk score subgroups.

RESULTS The total patient cohort comprised 811 patients with p16+ OPSCC (median age, 59.0 years [IQR, 47.4-70.6 years]; 683 men [84.2%]). In the external test set, the multiregion imaging risk score was found to be prognostic of DFS (HR, 3.76 [95% CI, 1.99-7.10]; $P < .001$), OS (HR, 4.80 [95% CI, 2.22-10.40]; $P < .001$), and LRF (HR, 4.47 [95% CI, 1.43-14.00]; $P = .01$) among all patients with p16+ OPSCC. The multiregion imaging risk score, integrating both PT and LN information, demonstrated a higher C index (0.63) compared with models focusing solely on PT (0.61) or LN (0.58). Chemotherapy was associated with improved DFS only among patients with high scores (HR, 0.09 [95% CI, 0.02-0.47]; $P = .004$) but not those with low scores (HR, 0.83 [95% CI, 0.32-2.10]; $P = .69$).

(continued)

Key Points

Question Can a deep learning model that uses both primary tumor and metastatic cervical lymph nodes from pretreatment computed tomography (CT) scans predict survival outcomes among patients with p16+ oropharyngeal squamous cell carcinoma (p16+ OPSCC)?

Findings In this prognostic study of 811 patients with p16+ OPSCC, a deep learning model integrating primary tumor and metastatic cervical lymph nodes significantly stratified patients into low- vs high-risk groups based on disease-free survival, overall survival, and locoregional failure. The model also identified candidates who would derive added benefit associated with concurrent chemoradiotherapy over definitive radiotherapy alone among patients with stage I disease.

Meaning These findings suggest that an imaging biomarker integrating patterns from both the primary tumor and the metastatic cervical lymph nodes from pretreatment CT scans is prognostic of survival outcomes and predictive of added benefit associated with chemotherapy for patients with p16+ OPSCC.

+ Supplemental content

Author affiliations and article information are listed at the end of this article.

Open Access. This is an open access article distributed under the terms of the CC-BY-NC-ND License, which does not permit alteration or commercial use, including those for text and data mining, AI training, and similar technologies.

Abstract (continued)

CONCLUSIONS AND RELEVANCE This prognostic study of p16+ OPSCC describes the development of a CT-based imaging risk score integrating PT and metastatic cervical LN features to predict recurrence risk and identify suitable candidates for treatment tailoring. This tool could optimize treatment modulations of p16+ OPSCC at a highly granular level.

JAMA Network Open. 2025;8(5):e258094. doi:10.1001/jamanetworkopen.2025.8094

Introduction

The increasing incidence of p16+ oropharyngeal squamous cell carcinoma (OPSCC) is resulting in a growing number of young survivors,¹ making posttreatment quality of life a major concern. Most p16+ OPSCC tumors tend to have a slower progression and may derive benefit associated with treatment deintensification.² However, previous efforts have shown that deintensification without careful patient selection can be detrimental.^{3,4} Better tools are needed to accurately identify patients with low- and intermediate-risk p16+ OPSCC who would truly derive benefit associated with reduced treatment intensity. Previous studies have demonstrated that chemotherapy can lead to heightened treatment-related toxic effects for a subset of patients with p16+ OPSCC with a low risk of disease recurrence.^{5,6} To minimize treatment toxic effects while maintaining survival outcomes among patients with p16+ OPSCC, recent studies have focused on identifying suitable candidates for treatment de-escalation, specifically by reducing the dose of chemoradiation without jeopardizing outcomes.^{3,7} This underscores the need for a better prognostic and predictive biomarker to identify patients with truly low-risk p16+ OPSCC who may not experience benefit associated with chemotherapy, allowing for a more optimized precision treatment strategy.

Presence of metastatic cervical lymph nodes (LNs) and tumor size are well-established prognostic factors for p16+ OPSCC,^{8,9} evaluated through TNM staging. T classification indicates the size of primary tumor (PT) and depth of invasion, while N classification denotes the size and the number of metastatic cervical LNs.¹⁰ Although TNM classification is effective for risk stratification in p16+ OPSCC, previous studies showed TNM classification is limited in quantifying intratumoral and intranodal heterogeneity.^{10,11} As the transition from the 7th to the 8th edition of the American Joint Committee on Cancer (AJCC) staging system highlights the prognostic significance of nodal classification,¹² there is a critical need to better characterize cervical nodal involvement to improve prognosis estimation in p16+ OPSCC.¹³

Radiologic computed tomographic (CT) imaging is routinely used for prognostication and therapeutic response assessment in p16+ OPSCC.¹⁴⁻¹⁶ The unique biology of the PT and its interaction with metastatic cervical LNs generate distinct imaging phenotypes. These phenotypes, when analyzed using deep learning algorithms, can be leveraged to predict clinical outcomes.^{17,18} Consequently, radiologic images combined with deep learning offer a noninvasive approach to identify tumor microenvironment characteristics relevant to LN metastases. Although considerable research has emphasized the prognostic value of the PT in radiographic imaging,¹⁹⁻²¹ comparatively less attention has been directed toward the combined artificial intelligence-based assessment of both the PT and cervical LN involvement.²²

In this study, we develop and validate a prognostic and predictive CT multiregion imaging risk score (SwinScore) integrating information from PT and metastatic LN features to predict p16+ OPSCC outcomes and benefit associated with chemotherapy. The multiregion imaging risk score was derived using hierarchical vision transformer architecture (Swin Transformer, version 2.0; Microsoft Corp; used June 2024)²³ as the backbone, which extracts features from both the PT and metastatic LNs. These features were then aggregated using cross-region self-attention²⁴ to enhance feature interactions across the 2 tumor habitat regions. In a multi-institutional cohort of 811 patients, we evaluated the multiregion imaging risk score as a prognostic biomarker for p16+ OPSCC. In addition,

we also investigated potential associations of multiregion imaging risk score subgroups with added benefit of chemotherapy in patients with AJCC 8th edition stage I and II disease.

Methods

Study Design, End Points, and CT Image Preprocessing

This retrospective analysis included 811 patients with p16+ OPSCC from 3 different sources: the Cancer Imaging Archive^{25,26} (TCIA; N = 522), which consists of the OPC (Oropharyngeal Carcinoma)-Radiomic cohort (n = 294), the HNSCC (Head and Neck Squamous Cell Carcinoma) cohort²⁷ (n = 134), the Head-Neck-PET [Positron Emission Tomography]-CT cohort²⁸ (n = 51), the HEAD-NECK-RADIOMICS-HN (Head Neck) 1 cohort²⁹ (n = 21), and the Cancer Genome Atlas (TCGA) cohort (n = 22). Patient data from TCIA were collected between the period from 1998 to 2013. The second source was collected from the Cleveland Clinic Foundation (CCF; N = 233), and the third source was obtained from the Winship Cancer Institute (Winship; N = 56). Patients were identified retrospectively within the period from 2002 to 2015 for the study. All patients included in this study went through definitive radiotherapy or concurrent chemoradiotherapy (eTable 1 in [Supplement 1](#)). This study was designed according to the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD) reporting guideline for a multivariable prediction model.³⁰ We randomly allocated 452 patients (86.6%) from the TCIA source to form the training set (D1) while the remaining 70 patients (13.4%) formed the internal validation set (D2). Patients from the CCF and Winship sources were combined to form the external test set (D3). A patient selection diagram with inclusion and exclusion criteria is provided in eFigure 1 in [Supplement 1](#). Clinical and outcome information of patients in the CCF and Winship sources were obtained by medical record review after approval from the Cleveland Clinic Foundation institutional review board and the Winship Cancer Institute institutional review board; because data were deidentified, the institutional review boards waived the requirement for written consent. Data from TCIA were deidentified and the requirement for written consent was waived.

All patients underwent an initial pretreatment CT scan with or without contrast agent injection for therapy planning (eMethods 1 and eTable 2 in [Supplement 1](#)). The manual segmentations were performed using the 3D Slicer software.³¹ The slices with dental artifacts were excluded for segmentations. Details on CT image preprocessing can be found in eMethods 2 in [Supplement 1](#).

Deep Learning Architecture, Training, and Evaluation

We designed our deep learning model using hierarchical vision transformer architecture²³ (eFigure 2 in [Supplement 1](#)) because it provides the flexibility of capturing both local and global information dependencies across PT and metastatic LNs. We used 4 hierarchical vision transformer blocks to separately extract features from the 2 cancer habitat regions, each containing distinct information. Within each hierarchical vision transformer block, we applied the self-attention mechanism²⁴ to enhance information interaction between PT and LN. We used the binary cross-entropy loss³² to supervise the learning process and output a continuous risk score (multiregion imaging risk score; median, -0.5; high, greater than -0.5; low, less than -0.5), which was trained in an end-to-end fashion using 3-year disease-free survival (DFS). More mathematical details of the model architecture, training details, and outcomes are provided in eMethods 3, 4, and 5 in [Supplement 1](#).

Statistical Analysis

Statistical analysis was performed between February and July 2024. The differences between the continuous variables (ie, age, smoking pack-year [PY], and multiregion imaging risk score) among datasets were compared using the 1-way analysis of variance test,³³ and the associations of the categorical variables were estimated using the χ^2 test.³⁴ The Harrell concordance index³⁵ (C index) was used as the performance metric. The median multiregion imaging risk score from D1 was chosen as the cutoff value to define high- and low-risk groups for plotting Kaplan-Meier³⁶ curves and

calculating the hazard ratios (HRs). Survival differences between multiregion imaging risk score groups were evaluated using the log-rank test.³⁷ The same Kaplan-Meier survival analyses were further performed within each cancer stage group defined by the AJCC 8th edition. To visualize individual variable importance in the multivariable Cox proportional hazards regression model, we computed the Shapley Additive exPlanations³⁸ (SHAP) values for all the variables considered in the multivariable analysis and visualized them with the bee swarm plot. To enhance model interpretability, we calculated the integrated gradients³⁹ of predictions and projected them onto the PT and the metastatic LNs, highlighting the areas contributing to the final predictions. We also quantified the associations of the multiregion imaging risk score's penultimate layer features ($n = 24$) with gene expression data for a subset of patients from TCGA (eMethods 6 in [Supplement 1](#)). Calibration curves along with the Hosmer-Lemeshow test⁴⁰ were used to compare the predicted survival probabilities with the actual probabilities. Decision curve analysis⁴¹ was performed to compare the net benefit associated with the multiregion imaging risk score, imaging risk scores derived from PT or LN alone, and a clinical model at various threshold probabilities. We investigated the potential association of the multiregion imaging risk score with chemotherapy benefit using an interaction test by adding an interaction term (multiregion imaging risk score $\times$ treatment) in the Cox proportional hazards regression model. Statistical analyses were performed using R, version 3.4.0 (R Project for Statistical Computing).⁴² All P values were from 2-tailed tests, and results were deemed statistically significant at $P < .05$.

Results

Patient Characteristics

Table 1 lists the clinicopathologic characteristics of the patients in D1 ($n = 452$), D2 ($n = 70$), and D3 ($n = 289$) cohorts. Of the 811 patients, 683 (84.2%) were male and 128 were female (15.8%), and the median age was 59.0 years (IQR, 47.4-70.6 years). The numbers of patients with stage I, II, and III diseases were 199 (24.5%), 435 (53.7%), and 177 (21.8%), respectively; 209 patients (25.8%) received definitive radiotherapy alone and 602 patients (74.2%) received concurrent chemoradiotherapy. The number of events for DFS was 222 (27.4%), for overall survival (OS) was 177 (21.8%), and for locoregional failure (LRF) was 64 (7.9%), with a median follow-up time of 65.9 months (IQR, 45.9-85.9 months).

Prognostic Values of Multiregion Imaging Risk Score Across and Within Stage Groups

The binary cross-entropy loss and C-indices with respect to epoch on D1 and D2 is provided in eFigure 3 in [Supplement 1](#). We divided patients into 2 risk groups using the median multiregion imaging risk score from D1 (-0.5). This cutoff point was then applied consistently across D2 and D3 for Kaplan-Meier survival analysis. Clinical characteristics within the high and low multiregion imaging risk score groups are summarized in eTable 3 in [Supplement 1](#). The prognostic values of the multiregion imaging risk score in predicting outcomes on D3 were consistently observed across the 3 end points (DFS: HR, 3.76 [95% CI, 1.99-7.10]; $P < .001$; C index, 0.63 [**Figure 1A**]; OS: HR, 4.80 [95% CI, 2.22-10.40]; $P < .001$; C index, 0.63 [**Figure 1B**]; LRF: HR, 4.47 [95% CI, 1.43-14.00; $P = .01$; C index, 0.63 [**Figure 1C**]). The multiregion imaging risk score, integrating both PT and LN information, demonstrated a higher C index (0.63) compared with models focusing solely on PT (0.61) or LN (0.58). The separation remained robust regardless of the chosen threshold cutoffs (eFigure 4 in [Supplement 1](#)). Kaplan-Meier analysis on D1 and D2 are provided in eFigure 5 in [Supplement 1](#). The subgroup analysis showed significant differences in DFS, OS, and LRF between the high and low multiregion imaging risk score groups within populations with AJCC 8th edition stage I disease (eFigure 6 in [Supplement 1](#)). The multiregion imaging risk score was not significantly associated with risk stratification of patients with AJCC 8th edition stage II disease (eFigure 7 in [Supplement 1](#)) but was significantly associated with risk stratification of patients with AJCC 8th edition stage III disease for DFS and OS (eFigure 8 in [Supplement 1](#)). Calibration curves showed good agreement between

the multiregion imaging risk score predictions and the actual survival rates (eFigure 9 in Supplement 1).

Multivariable Cox proportional hazards regression (Table 2) confirmed the multiregion imaging risk score as an independent prognostic factor for DFS (HR, 2.08 [95% CI, 1.30-3.33]; $P = .002$) and OS (HR, 3.58 [95% CI, 1.96-4.53] $P = .003$). SHAP values ranked the multiregion imaging risk score as the most important variable in the model (eFigure 10 in Supplement 1; Figure 1D).

Prognostic Value of PT and Metastatic LNs

We fed either PT or LN data alone into the model and obtained corresponding risk scores, which we termed as *SwinPT* and *SwinLN*, respectively. The multiregion imaging risk score achieved the highest C index for DFS and OS predictions but was inferior to *SwinPT* for LRF prediction (eFigure 11 in Supplement 1), suggesting that while the PT is essential for LRF detection, metastatic cervical LN data add prognostic value for survival outcomes (ie, DFS, OS). We conducted a decision curve analysis and found that the multiregion imaging risk score consistently outperformed *SwinPT*,

Table 1. Clinicopathologic and Demographic Characteristics of the Patients

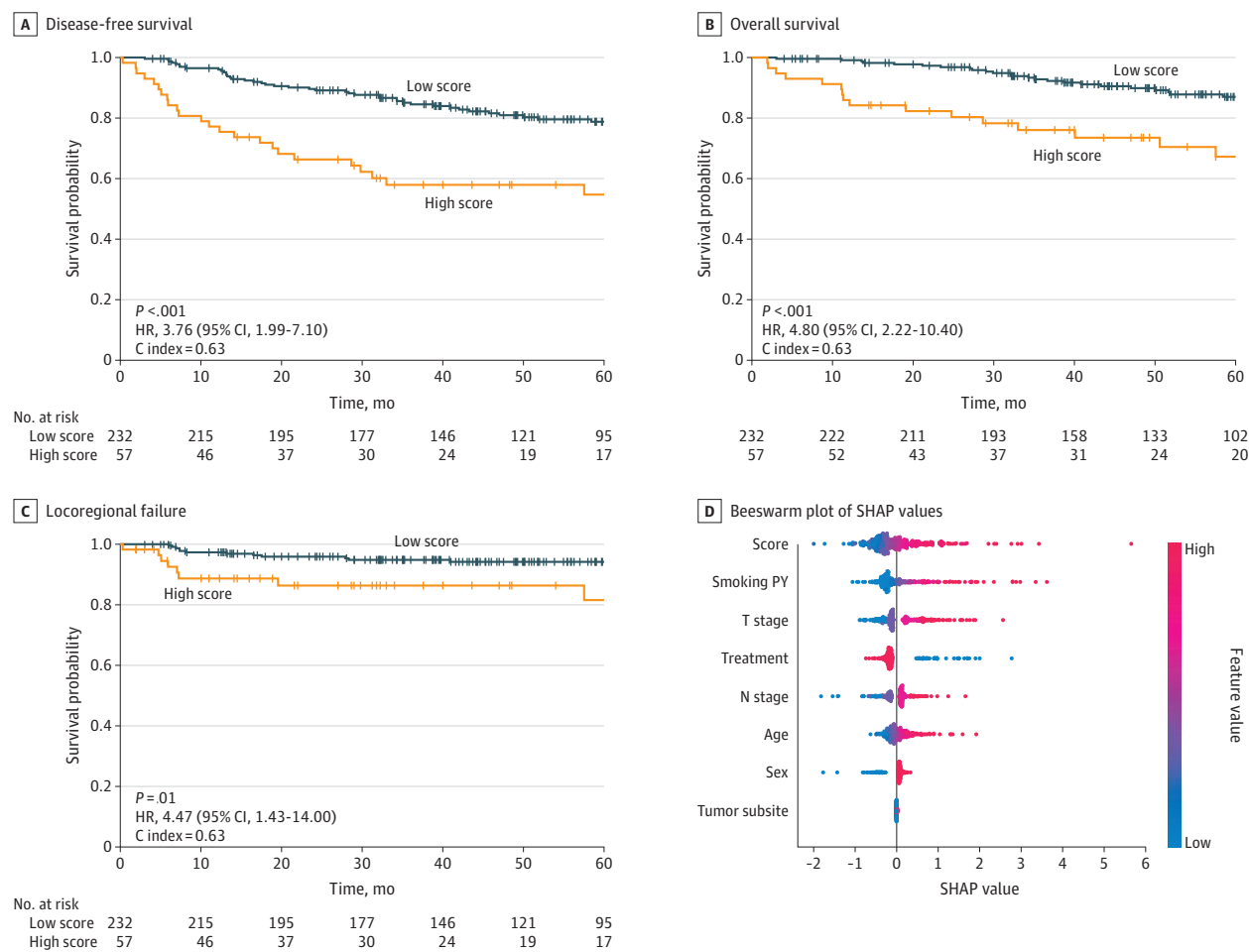
Characteristic	Training set (n = 452)	Internal validation set (n = 70)	External test set (n = 289)
Age, mean (SD), y	58.8 (12.4)	59.9 (11.7)	59.6 (12.2)
Smoking, mean (SD), pack-years	15.0 (21.7)	13.0 (22.6)	15.2 (20.4)
Sex, No. (%)			
Male	374 (82.7)	52 (74.3)	257 (88.9)
Female	78 (17.3)	18 (25.7)	32 (11.1)
N stage, No. (%)			
N0	59 (13.1)	13 (18.6)	16 (5.5)
N1	75 (16.6)	8 (11.4)	107 (37.0)
N2	294 (65.0)	45 (64.3)	143 (49.5)
N3	24 (5.4)	4 (5.7)	23 (8.0)
T stage, No. (%)			
T1	109 (24.1)	7 (10.0)	73 (25.2)
T2	169 (37.4)	32 (45.7)	112 (38.8)
T3	114 (25.2)	14 (20.0)	47 (16.3)
T4	60 (13.3)	17 (24.3)	57 (19.7)
AJCC 8th edition stage, No. (%)			
I	95 (21.0)	13 (18.6)	91 (31.5)
II	279 (61.7)	36 (51.4)	120 (41.5)
III	78 (17.3)	21 (30.0)	78 (27.0)
Treatment, No. (%)			
Radiotherapy	162 (35.8)	25 (35.7)	22 (7.6)
Chemoradiotherapy	290 (64.2)	45 (64.3)	267 (92.4)
Tumor subsite, No. (%)			
Base of tongue	181 (40.0)	46 (65.7)	187 (64.7)
Tonsil	228 (50.4)	23 (32.9)	99 (34.3)
Posterior wall	43 (9.5)	1 (1.4)	3 (1.0)
Disease-free survival, No. (%)			
Event	134 (29.6)	18 (25.7)	70 (24.2)
Censored	318 (70.4)	52 (74.3)	219 (75.8)
Overall survival, No. (%)			
Event	117 (25.9)	14 (20.0)	46 (15.9)
Censored	335 (74.1)	56 (80.0)	243 (84.1)
Locoregional failure, No. (%)			
Event	39 (8.6)	4 (5.7)	21 (7.3)
Censored	413 (91.4)	66 (94.3)	268 (92.7)

Abbreviation: AJCC, American Joint Committee on Cancer.

SwinLN, and a clinical model (T stage, N stage, and smoking pack-year) in predicting DFS (eFigure 12B in Supplement 1) and OS (eFigure 12C in Supplement 1) across threshold probabilities, while the multiregion imaging risk and SwinPT showed comparable net benefit associated with LRF prediction (eFigure 12D in Supplement 1). The integrated gradient attention maps on CT images aligned well with the regions corresponding to PT and metastatic cervical LNs (Figure 2). To study potential associations of the imaging risk score (ie, the multiregion imaging risk score and imaging risk scores derived from PT or LN alone) with clinical factors, we calculated the point-biserial correlation coefficients (eTable 4 in Supplement 1). We also compared the multiregion imaging risk score against 17 state-of-the-art models with various feature extraction backbones (ie, radiomic,¹⁵ convolutional neural network,¹⁸ vision transformer,⁴³ and hierarchical vision transformer²³), cancer habitat regions (ie, PT and LN), and fusion strategies (ie, concatenation, Kronecker product,⁴⁴ and self attention²⁴). The multiregion imaging risk score outperformed all models in predicting DFS on D3 (eTable 5 in Supplement 1).

Correlation analysis and gene set enrichment of the multiregion imaging risk score's penultimate layer features revealed significant immune-related pathways and gene sets (eFigure 13 in Supplement 1), highlighting their importance in patient stratification and providing a potential biological basis for the multiregion imaging risk score method.

Figure 1. Prognostic Value of the Multiregion Imaging Risk Score



A, Kaplan-Meier survival analysis of the score on the external test set using disease-free survival as the end point. B, Kaplan-Meier survival analysis of the score on the external test set using overall survival as the end point. C, Kaplan-Meier survival analysis of the score on the external test set using locoregional failure as the end point. D, Beeswarm

plot of Shapley Additive exPlanations (SHAP) variable importance in multivariate Cox proportional hazards regression analysis of disease-free survival. C index indicates concordance index; HR, hazard ratio; and PY, pack-years.

Association of Multiregion Imaging Risk Score With Chemotherapy Benefit

We further assessed the multiregion imaging risk’s predictive value by investigating if dichotomized multiregion imaging risk score subgroups were associated with chemotherapy benefit. This analysis focused primarily on patients with AJCC 8th edition stage I and stage II tumors, where an individualized decision regarding chemotherapy would be the most relevant.^{15,45} Clinical characteristics and survival rates according to treatment arms are provided in eTable 6 and eFigure 14 in Supplement 1. For patients in the high multiregion imaging risk score group, the 3-year DFS death or failure rate was significantly lower with chemoradiotherapy compared with radiotherapy alone (D1: 18% vs 52%; χ^2 test $P = .04$; and D2 + D3: 24% vs 72%; χ^2 test $P = .01$), while those in the low multiregion imaging risk score group showed minimal differences between treatments (eFigure 15 in Supplement 1). These results suggest that patients with high scores may derive more benefit associated with chemotherapy, as indicated by their lower event rates when chemotherapy was added to radiotherapy.

Comparison of DFS between treatment arms showed that patients with high scores experienced significant benefit associated with chemotherapy vs radiotherapy (D1: HR, 0.50 [95% CI, 0.31-0.80]; $P = .004$; D2 + D3: HR, 0.09 [95% CI, 0.02-0.47]; $P = .004$), while those with low scores did not (D1: HR, 0.32 [95% CI, 0.14-3.83]; $P = .31$; D2 + D3: HR, 0.83 [95% CI, 0.32-2.10]; $P = .69$) (Figure 3). A significant interaction between the multiregion imaging risk score and chemotherapy was confirmed, highlighting the differential treatment benefit based on the multiregion imaging risk score.

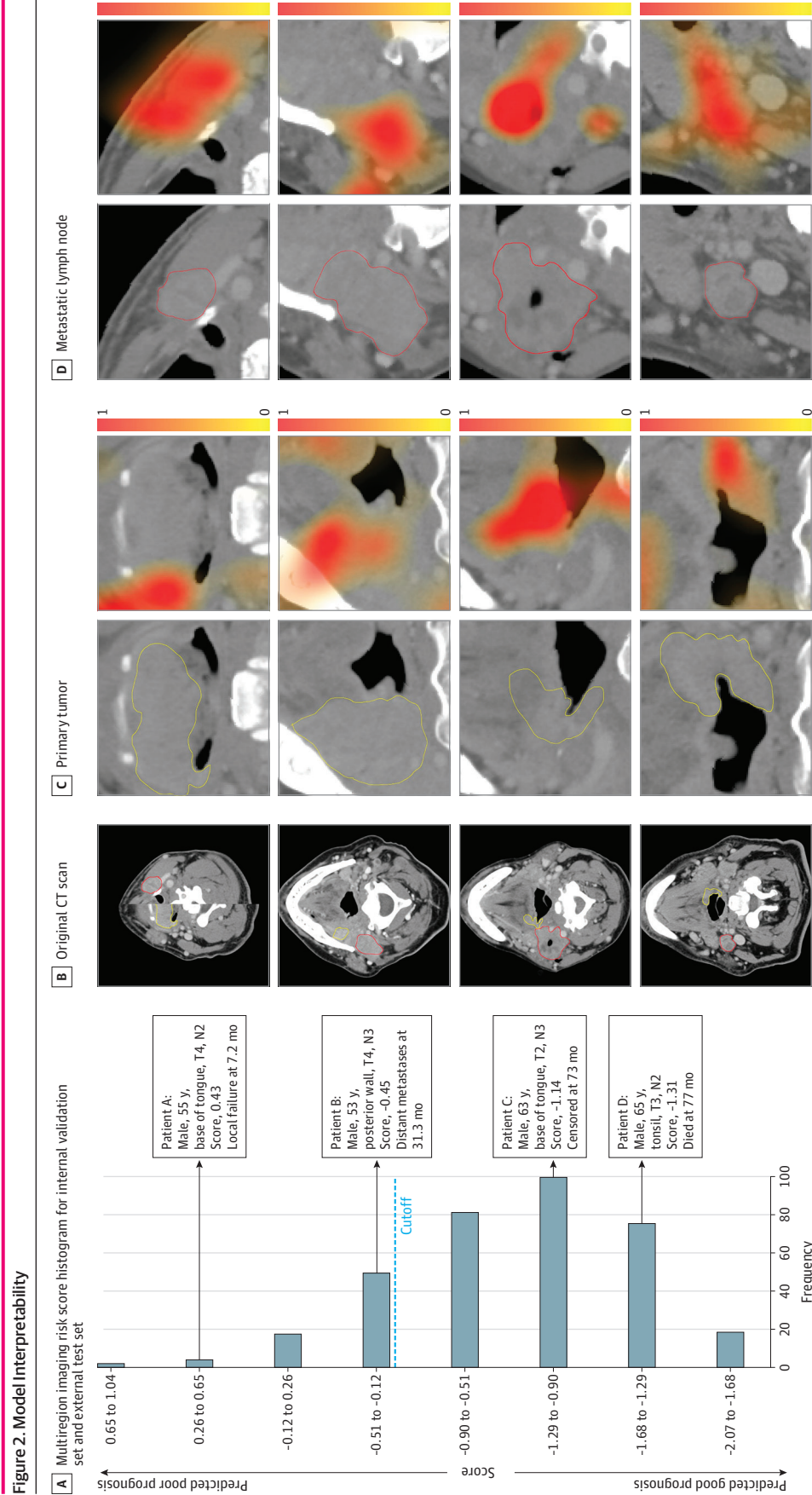
Discussion

Given the generally favorable prognosis of patients with p16+ OPSCC, considerable efforts have been directed toward developing precision oncology strategies, including the targeted reduction of radiotherapy and chemotherapy doses and schedules.^{5,46} However, the variable clinical behavior of

Table 2. Multivariable Cox Proportional Hazards Regression Analysis on Disease-Free Survival and Overall Survival in the External Test Set

Parameter	Disease-free survival		Overall survival	
	HR (95% CI)	P value	HR (95% CI)	P value
Age	1.02 (0.99-1.05)	.20	1.03 (1.01-1.07)	.02 ^a
Sex				
Female	1 [Reference]	NA	1 [Reference]	NA
Male	1.49 (0.64-3.50)	.35	1.67 (0.70-3.98)	.25
Smoking pack-years	1.02 (1.01-1.03)	<.001 ^a	1.02 (1.01-1.04)	<.001 ^a
Tumor subsite				
Base of tongue	1 [Reference]	NA	1 [Reference]	NA
Posterior wall or soft palate	4.30 (1.20-15.40)	.03 ^a	2.29 (0.87-5.40)	.42
Tonsil	0.97 (0.56-1.68)	.91	0.89 (0.49-1.63)	.71
T stage				
T1	1 [Reference]	NA	1 [Reference]	NA
T2	1.05 (0.54-2.01)	.90	0.96 (0.44-2.09)	.92
T3	1.15 (0.51-2.57)	.74	1.72 (0.53-4.67)	.36
T4	1.75 (0.84-3.66)	.14	1.73 (1.27-3.89)	.003 ^a
N stage				
N0	1 [Reference]	NA	1 [Reference]	NA
N1	0.72 (0.27-1.95)	.52	2.61 (0.45-5.66)	.54
N2	0.80 (0.31-2.10)	.65	1.45 (0.21-3.30)	.39
N3	1.40 (0.44-4.49)	.57	3.64 (0.32-5.24)	.56
Multiregion imaging risk score (continuous)	2.08 (1.30-3.33)	.002 ^a	3.58 (1.96-4.53)	.003 ^a

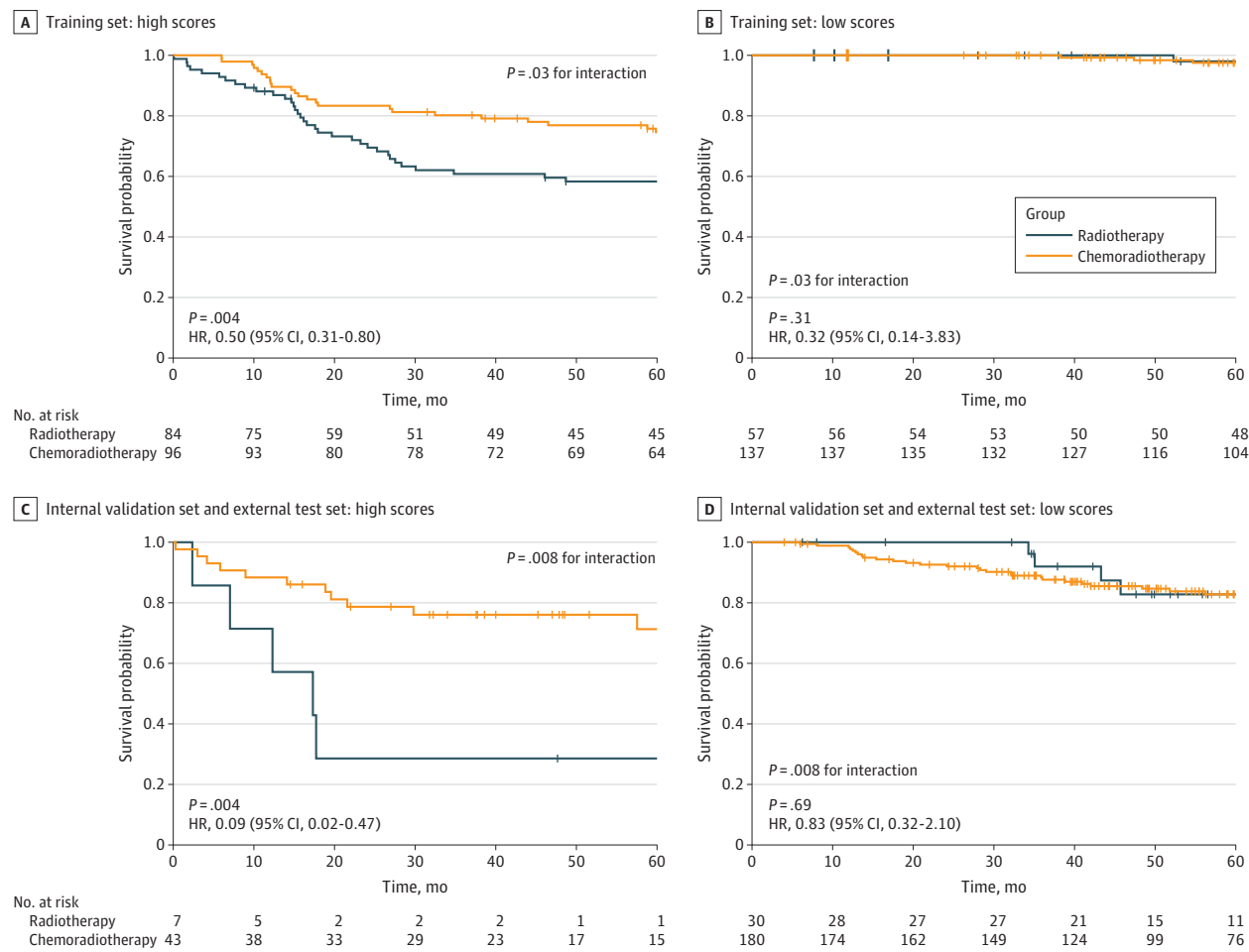
Abbreviations: HR, hazard ratio; NA, not applicable.
^a Statistically significant by 2-tailed test at $P < .05$.



p16+ OPSCC poses a significant risk of overtreatment or undertreatment for some patients.^{47,48} Although a significant portion of patients with p16+ OPSCC achieve a cure with concurrent chemoradiotherapy, they often experience significant toxic effects and morbidity.^{49,50} This outcome has driven oncology research communities to identify low-risk patients who might experience benefit associated with de-escalated treatment to minimize the acute and long-term adverse effects of chemoradiotherapy without affecting cure rates.⁵¹ In the NRG-HN002 trial reported by Yom et al,³ patients with low-risk human papillomavirus-associated OPSCC were randomly assigned to either dose-reduced radiotherapy with weekly cisplatin or accelerated radiotherapy (60 Gy) alone. A portion of patients in this trial developed local-regional failure and/or distant metastasis, indicating that the low-risk criteria used were not adequately precise to discern patients who could derive benefit associated with treatment de-escalation.⁵² As such, accurate risk stratification and the identification of patients who are most likely to experience benefit associated with treatment intensification are crucial for providing precision care for patients with p16+ OPSCC.

In this work, we developed a prognostic and predictive CT-based biomarker, a multiregion imaging risk score, to predict survival outcomes and added benefit associated with chemotherapy

Figure 3. Multiregion Imaging Risk Score as a Predictive Biomarker of Added Benefit Associated With Chemotherapy



A, Kaplan-Meier survival analysis of disease-free survival (DFS) between treatment arms on the training set for patients with high scores. B, Kaplan-Meier survival analysis of DFS between treatment arms on the training set for patients with low scores. C, Kaplan-Meier survival analysis of DFS between treatment arms on the internal validation set and the external test set for patients with high scores. D, Kaplan-Meier survival analysis of DFS between treatment arms on the internal validation set and the external test set for

patients with low scores. Patients with high scores who received chemotherapy after radiotherapy experienced a significant DFS benefit compared with patients receiving radiotherapy alone, while patients with low scores did not ($P = .03$ for interaction for the training set; $P = .008$ for interaction for the internal validation set and the external test set). HR indicates hazard ratio.

using a cohort of 811 patients with p16+ OPSCC. This multiregion imaging risk score, calculated using a hierarchical vision transformer²³ model that integrates spatial data from both PT and metastatic cervical LNs, significantly stratified patients into 2 distinct risk groups and was an independent prognostic marker for OS and DFS. In addition, the multiregion imaging risk score was found to be strongly associated with outcomes even within the AJCC 8th edition's definition of patients with stage I disease, who are currently recommended for de-escalation treatment.¹⁰ If validated in a prospective clinical trial, this multiregion imaging risk score could significantly impact clinical practice by identifying high-risk patients with stage I disease and enhancing outcomes for genuinely low-risk patients suitable for more refined treatment tailoring.

A distinctive aspect of our study is that the imaging patterns of metastatic cervical LNs and their interactions with the PT were automatically quantified via deep learning, the self-attention mechanism,²⁴ and the Kronecker product. Previous studies typically extracted features from only the PT or combined hand-crafted radiomic features from both PT and LNs, followed by Cox proportional hazards regression to select representative features, which overlooks interactions between tumor habitats regions.^{16,53,54} In contrast, the multiregion imaging risk score captures region-wise interactions between the PT and LNs using self-attention after each embedding downscale within the hierarchical vision transformer blocks, allowing prognostic features to be learned from both regions simultaneously. The premise behind such fusion schemes is that the intricate feature interplays among PT and metastatic cervical LNs may have been overlooked during more naive fusion strategies, such as concatenation.⁵⁵ From the results (Figure 3) we can clearly observe that metastatic LNs add value in DFS and OS prediction but not in LRF detection. This finding could enhance our understanding of the prognostic significance of metastatic cervical LNs, suggesting they may be more closely associated with survival outcomes rather than LRF.

Besides studying the prognostic value of the multiregion imaging risk score, we explored its potential association with treatment outcomes in populations with AJCC 8th edition stage I and II p16+ OPSCC. Our findings suggest that the multiregion imaging risk score may provide alternative evidence to support treatment intensification strategies. Although prior research concentrated on OPSCC risk stratification,⁵⁶⁻⁵⁹ these studies failed to deliver personalized treatment strategies based solely on the risk profile. In our study, we found that patients with high scores with stage I and II tumors exhibited a lower hazard of death or treatment failure when treated with chemotherapy, as measured by DFS. In contrast, patients with low scores showed nonsignificant HRs between the 2 treatment arms when using DFS as the end point. Formal interaction tests were performed to prove there exists a joint association with survival induced by both treatment as well as the multiregion imaging risk score. However, among patients with stage II disease, the differences in outcomes (DFS, OS, LRF) between risk groups did not reach statistical significance in the validation cohorts (eFigure 7 in Supplement 1). These findings suggest that the multiregion imaging risk score's predictive utility is most robust in stage I disease, where it may help identify candidates likely to derive survival benefit associated with chemotherapy. For patients with stage II disease, the lack of a statistically significant prognostic value in validation cohorts limits definitive conclusions.

Limitations

Our study had several limitations. First, the deep learning approach in this study carries less interpretability compared with conventional radiomic approaches, in which each feature possesses specific physical meanings (eg, tumor shape or texture patterns). Second, only the largest metastatic cervical LN, if available, was analyzed for each patient, which means the prognostic value coming from LNs might not be fully exploited. Third, this study is retrospective and therefore susceptible to confounding factors and biases (eg, radiation doses and type of chemotherapy were not strictly controlled). The lack of granular chemotherapy- and radiotherapy-related details in the TCIA datasets (D1 and D2) raises uncertainty about whether the multiregion imaging risk score's prognostic and predictive performance generalizes to homogeneous treatment protocols. Although these findings are hypothesis generating, they should not be interpreted as definitive evidence of

the multiregion imaging risk score's predictive utility for chemotherapy benefit. Instead, the score's association with treatment outcomes may serve as supplementary evidence to inform treatment intensification decisions, particularly in the context of stage I p16+ OPSCC. The multiregion imaging risk score needs to be validated in uniformly treated retrospective and prospective clinical trial cohorts to confirm its clinical utility and generate level 1b and 1a evidence, respectively.

Conclusions

In this prognostic study, we developed and validated a prognostic and predictive biomarker, a multiregion imaging risk score, which allows p16+ OPSCC risk stratification by mining features from PT and metastatic cervical LNs on routinely acquired CT scans. Within AJCC 8th edition stage I tumors, the multiregion imaging risk score also seems to be predictive of added benefit associated with chemotherapy, allowing the community to optimize treatment modulations of p16+ OPSCC at a highly granular level. These advancements highlight the score's potential to refine personalized therapeutic strategies, reduce overtreatment, and improve clinical decision-making in OPSCC.

ARTICLE INFORMATION

Accepted for Publication: February 28, 2025.

Published: May 1, 2025. doi:10.1001/jamanetworkopen.2025.8094

Open Access: This is an open access article distributed under the terms of the [CC-BY-NC-ND License](#), which does not permit alteration or commercial use, including those for text and data mining, AI training, and similar technologies. © 2025 Song B et al. *JAMA Network Open*.

Corresponding Author: Anant Madabhushi, PhD, Emory University, 1750 Haygood Dr, Ste N647, Atlanta, GA 30322 (anantm@emory.edu).

Author Affiliations: Wallace H. Coulter Department of Biomedical Engineering, Georgia Institute of Technology and Emory University, Atlanta, Georgia (Song, Khalighi, Pandav, Dam, Sonuga, Madabhushi); Therapanacea, Paris, France (Leroy); Department of Radiation Oncology, Holden Comprehensive Cancer Center, Iowa Neuroscience Institute, University of Iowa, Iowa City (Yang, Koyfman); Diagnostics Institute, Cleveland Clinic, Cleveland, Ohio (Lee, Stock); Department of Radiology and Imaging Sciences, Emory University Hospital, Atlanta, Georgia (Li); Department of Population and Quantitative Health Sciences, Case Western Reserve University, Cleveland, Ohio (Fu); Department of Hematology and Medical Oncology, Winship Cancer Institute, Atlanta, Georgia (Saba); Department of Otolaryngology, Winship Cancer Institute, Atlanta, Georgia (Patel); Atlanta Veterans Affairs Medical Center, Atlanta, Georgia (Madabhushi).

Author Contributions: Dr Song had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Song, Koyfman, Madabhushi.

Acquisition, analysis, or interpretation of data: Song, Leroy, Yang, Khalighi, Pandav, Dam, Lee, Stock, Li, Sonuga, Fu, Saba, Patel, Madabhushi.

Drafting of the manuscript: Song, Leroy, Yang, Khalighi, Saba.

Critical review of the manuscript for important intellectual content: Song, Yang, Khalighi, Pandav, Dam, Lee, Stock, Li, Sonuga, Fu, Koyfman, Saba, Patel, Madabhushi.

Statistical analysis: Song, Leroy, Yang, Khalighi, Dam.

Obtained funding: Madabhushi.

Administrative, technical, or material support: Pandav, Sonuga, Madabhushi.

Supervision: Lee, Li, Fu, Madabhushi.

Conflict of Interest Disclosures: Dr Leroy reported receiving grants from Therapanacea during the conduct of the study. Dr Fu reported receiving grants from Case Western Reserve University during the conduct of the study. Dr Koyfman reported receiving personal fees from Merck, BMS, Regeneron, Castle Biosciences, Galera Therapeutics, UpToDate, and Varian Medical Systems outside the submitted work. Dr Saba reported receiving personal fees from Astra Zeneca, Eisai Medical, Exelixis, Merck, EMD Serono, Pfizer, Kura, Vaccinex, Cue Biopharma, BionTech, GSK, Tosk, Seagen, Flamingo, Infinity, Inovio, Aveo, Medscape, Onclive, UpToDate, BMS, Cornerstone, Celldex, Surface

Oncology, Astex, Imugene, Faron Pharmaceutical, Coherus, Adagene, Fulgent, Springer, Nanobiotix, and Taiho outside the submitted work and funding from BMS and Exelixis. Dr Patel reported receiving personal fees from Intuitive Surgical outside the submitted work. Prof Madabhushi reported being co-founder of and holding equity in Picture Health, Elucid Bioimaging, and Inspirata Inc; receiving grants from Bristol Myers-Squibb and AstraZeneca during the conduct of the study; and receiving personal fees from Simbiosys Inc, Takeda Inc, Aiforia Inc, Caris Inc, and Roche Inc outside the submitted work; in addition, Prof Madabhushi had a patent for USSN 9,767,555 issued. No other disclosures were reported.

Funding/Support: This work was supported by the National Cancer Institute under awards R01CA268287A1, U01CA269181, R01CA26820701A1, R01CA249992-01A1, R01CA202752-01A1, R01CA208236-01A1, R01CA216579-01A1, R01CA220581-01A1, R01CA257612-01A1, 1U01CA239055-01, 1U01CA248226-01, and 1U54CA254566-01; the National Heart, Lung, and Blood Institute under awards 1R01HL15127701A1 and R01HL15807101A1; the National Institute of Biomedical Imaging and Bioengineering under award 1R43EB028736-01; Veterans Affairs Merit Review Award IBX004121A from the US Department of Veterans Affairs Biomedical Laboratory Research and Development Service the Office of the Assistant Secretary of Defense for Health Affairs; through the Breast Cancer Research Program (award W81XWH-19-1-0668); the Prostate Cancer Research Program (award W81XWH-20-1-0851); the Lung Cancer Research Program (awards W81XWH-18-1-0440 and W81XWH-20-1-0595); the Peer Reviewed Cancer Research Program (awards W81XWH-18-1-0404, W81XWH-21-1-0345, and W81XWH-21-1-0160); the Kidney Precision Medicine Project Glue Grant; and sponsored research agreements from Bristol Myers Squibb, Boehringer-Ingelheim, Eli Lilly, and AstraZeneca.

Role of the Funder/Sponsor: The funding sources had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Disclaimer: The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health, the US Department of Veterans Affairs, the Department of Defense, or the United States Government.

Data Sharing Statement: See [Supplement 2](#).

REFERENCES

1. Amin DR, Philips R, Bertoni DG, et al. Differences in functional and survival outcomes between patients receiving primary surgery vs chemoradiation therapy for treatment of T1-T2 oropharyngeal squamous cell carcinoma. *JAMA Otolaryngol Head Neck Surg*. 2023;149(11):980-986. doi:10.1001/jamaoto.2023.1944
2. Adelstein DJ, Ismaila N, Ku JA, et al. Role of treatment deintensification in the management of p16+ oropharyngeal cancer: ASCO provisional clinical opinion. *J Clin Oncol*. 2019;37(18):1578-1589. doi:10.1200/JCO.19.00441
3. Yom SS, Torres-Saavedra P, Caudell JJ, et al. Reduced-dose radiation therapy for HPV-associated oropharyngeal carcinoma (NRG Oncology HNO02). *J Clin Oncol*. 2021;39(9):956-965. doi:10.1200/JCO.20.03128
4. Saba NF, Pamulapati S, Patel B, et al. Novel immunotherapeutic approaches to treating HPV-related head and neck cancer. *Cancers (Basel)*. 2023;15(7):1959. doi:10.3390/cancers15071959
5. Skillington SA, Kallogjeri D, Lewis JS Jr, Piccirillo JF. The role of adjuvant chemotherapy in surgically managed, p16-positive oropharyngeal squamous cell carcinoma. *JAMA Otolaryngol Head Neck Surg*. 2017;143(3):253-259. doi:10.1001/jamaoto.2016.3353
6. Hargreaves S, Beasley M, Hurt C, Jones TM, Evans M. Deintensification of adjuvant treatment after transoral surgery in patients with human papillomavirus-positive oropharyngeal cancer: the conception of the PATHOS study and its development. *Front Oncol*. 2019;9:936. doi:10.3389/fonc.2019.00936
7. Chera BS, Amdur RJ, Green R, et al. Phase II trial of de-intensified chemoradiotherapy for human papillomavirus-associated oropharyngeal squamous cell carcinoma. *J Clin Oncol*. 2019;37(29):2661-2669. doi:10.1200/JCO.19.01007
8. Vijayvargiya P, Trivedi S, Rupji M, et al. Comparison of the seventh and eighth edition of American Joint Committee on Cancer (AJCC) Staging for selected and nonselected oropharyngeal squamous cell carcinomas. *Oncologist*. 2022;27(1):48-56. doi:10.1093/oncolo/oyab001
9. Lechner M, Liu J, Masterson L, Fenton TR. HPV-associated oropharyngeal cancer: epidemiology, molecular biology and clinical management. *Nat Rev Clin Oncol*. 2022;19(5):306-327. doi:10.1038/s41571-022-00603-7
10. Machczyński P, Majchrzak E, Niewinski P, Marchlewska J, Goluśński W. A review of the 8th edition of the AJCC staging system for oropharyngeal cancer according to HPV status. *Eur Arch Otorhinolaryngol*. 2020;277(9):2407-2412. doi:10.1007/s00405-020-05979-9

11. van Gysen K, Stevens M, Guo L, et al. Validation of the 8th edition UICC/AJCC TNM staging system for HPV associated oropharyngeal cancer patients managed with contemporary chemo-radiotherapy. *BMC Cancer*. 2019; 19(1):674. doi:[10.1186/s12885-019-5894-8](https://doi.org/10.1186/s12885-019-5894-8)
12. Huang SH, O'Sullivan B, Su J, et al. Prognostic importance of radiologic extranodal extension in HPV-positive oropharyngeal carcinoma and its potential role in refining TNM-8 cN-classification. *Radiother Oncol*. 2020; 144:13-22. doi:[10.1016/j.radonc.2019.10.011](https://doi.org/10.1016/j.radonc.2019.10.011)
13. Xiao R, Ward MC, Yang K, et al. The prognostic impact of level I lymph node involvement in oropharyngeal squamous cell carcinoma. *Head Neck*. 2019;41(11):3895-3905. doi:[10.1002/hed.25927](https://doi.org/10.1002/hed.25927)
14. Subramaniam RM, Alluri KC, Tahari AK, Aygun N, Quon H. PET/CT imaging and human papilloma virus-positive oropharyngeal squamous cell cancer: evolving clinical imaging paradigm. *J Nucl Med*. 2014;55(3):431-438. doi:[10.2967/jnumed.113.125542](https://doi.org/10.2967/jnumed.113.125542)
15. Song B, Yang K, Viswanathan VS, et al. CT radiomic signature predicts survival and chemotherapy benefit in stage I and II HPV-associated oropharyngeal carcinoma. *NPJ Precis Oncol*. 2023;7(1):53. doi:[10.1038/s41698-023-00404-w](https://doi.org/10.1038/s41698-023-00404-w)
16. Haider SP, Zeevi T, Baumeister P, et al. Potential added value of PET/CT radiomics for survival prognostication beyond AJCC 8th edition staging in oropharyngeal squamous cell carcinoma. *Cancers (Basel)*. 2020;12(7):1778. doi:[10.3390/cancers12071778](https://doi.org/10.3390/cancers12071778)
17. Lan T, Kuang S, Liang P, et al. MRI-based deep learning and radiomics for prediction of occult cervical lymph node metastasis and prognosis in early-stage oral and oropharyngeal squamous cell carcinoma: a diagnostic study. *Int J Surg*. 2024;110(8):4648-4659. doi:[10.1097/JS9.0000000000001578](https://doi.org/10.1097/JS9.0000000000001578)
18. Cheng NM, Yao J, Cai J, et al. Deep learning for fully automated prediction of overall survival in patients with oropharyngeal cancer using FDG-PET imaging. *Clin Cancer Res*. 2021;27(14):3948-3959. doi:[10.1158/1078-0432.CCR-20-4935](https://doi.org/10.1158/1078-0432.CCR-20-4935)
19. Song B, Yang K, Garneau J, et al. Radiomic features associated with HPV status on pretreatment computed tomography in oropharyngeal squamous cell carcinoma inform clinical prognosis. *Front Oncol*. 2021;11:744250. doi:[10.3389/fonc.2021.744250](https://doi.org/10.3389/fonc.2021.744250)
20. Ma B, Guo J, Zhai TT, et al. CT-based deep multi-label learning prediction model for outcome in patients with oropharyngeal squamous cell carcinoma. *Med Phys*. 2023;50(10):6190-6200. doi:[10.1002/mp.16465](https://doi.org/10.1002/mp.16465)
21. De Biase A, Ma B, Guo J, et al. Deep learning-based outcome prediction using PET/CT and automatically predicted probability maps of primary tumor in patients with oropharyngeal cancer. *Comput Methods Programs Biomed*. 2024;244:107939. doi:[10.1016/j.cmpb.2023.107939](https://doi.org/10.1016/j.cmpb.2023.107939)
22. Wu J, Gensheimer MF, Zhang N, et al. Integrating tumor and nodal imaging characteristics at baseline and mid-treatment computed tomography scans to predict distant metastasis in oropharyngeal cancer treated with concurrent chemoradiotherapy. *Int J Radiat Oncol Biol Phys*. 2019;104(4):942-952. doi:[10.1016/j.ijrobp.2019.03.036](https://doi.org/10.1016/j.ijrobp.2019.03.036)
23. Liu Z, Lin Y, Cao Y, et al. Swin Transformer: hierarchical vision transformer using shifted windows. Paper presented at: IEEE/CVF International Conference on Computer Vision (ICCV); October 10-17, 2021; Montreal, Quebec, Canada: 9992-10002. Accessed March 20, 2025. https://openaccess.thecvf.com/content/ICCV2021/html/Liu_Swin_Transformer_Hierarchical_Vision_Transformer_Using_Shifted_Windows_ICCV_2021_paper
24. Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need. *arXiv*. Preprint posted online June 12, 2017. Last revised August 2, 2023. doi:[10.48550/arXiv.1706.03762](https://doi.org/10.48550/arXiv.1706.03762)
25. Kwan JYY, Su J, Huang SH, et al. Radiomic biomarkers to refine risk models for distant metastasis in HPV-related oropharyngeal carcinoma. *Int J Radiat Oncol Biol Phys*. 2018;102(4):1107-1116. doi:[10.1016/j.ijrobp.2018.01.057](https://doi.org/10.1016/j.ijrobp.2018.01.057)
26. Clark K, Vendt B, Smith K, et al. The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository. *J Digit Imaging*. 2013;26(6):1045-1057. doi:[10.1007/s10278-013-9622-7](https://doi.org/10.1007/s10278-013-9622-7)
27. Grossberg A, Elhalawani H, Mohamed A, et al; M.D. Anderson Cancer Center Head and Neck Quantitative Imaging Working Group. HNSCC version 4 [dataset]. The Cancer Imaging Archive. 2020. Accessed March 20, 2025. <https://doi.org/10.7937/k9/tcia.2020.a8sh-7363>
28. Vallières M, Kay-Rivest E, Perrin LJ, et al. Data from Head-Neck-PET-CT. The Cancer Imaging Archive. 2017. Accessed March 20, 2025. <https://doi.org/10.7937/K9/TCIA.2017.8oje5q00>
29. Wee L, Dekker A. Data from HEAD-NECK-RADIOMICS-HN1 [data set]. The Cancer Imaging Archive. 2019. Accessed March 20, 2025. <https://www.cancerimagingarchive.net/collection/head-neck-radiomics-hn1/>

30. Collins GS, Reitsma JB, Altman DG, Moons KG. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD Statement. *BMC Med*. 2015;13:1. doi:10.1186/s12916-014-0241-z
31. Fedorov A, Beichel R, Kalpathy-Cramer J, et al. 3D Slicer as an image computing platform for the Quantitative Imaging Network. *Magn Reson Imaging*. 2012;30(9):1323-1341. doi:10.1016/j.mri.2012.05.001
32. Topsøe F. Bounds for entropy and divergence for distributions over a two-element set. *JIPAM J Inequalities Pure Appl Math*. 2001;2(2):Paper No. 25, 13 p. Accessed March 20, 2025. <https://eudml.org/doc/122035>
33. Welch BL. On the comparison of several mean values: an alternative approach. *Biometrika*. 1951;38(3/4):330-336. doi:10.2307/2332579
34. Cochran WG. The χ^2 test of goodness of fit. *Ann Math Stat*. 1952;23(3):315-345. doi:10.1214/aoms/117729380
35. Koziol JA, Jia Z. The concordance index C and the Mann-Whitney parameter $\Pr(X>Y)$ with randomly censored data. *Biom J*. 2009;51(3):467-474. doi:10.1002/bimj.200800228
36. Stel VS, Dekker FW, Tripepi G, Zoccali C, Jager KJ. Survival analysis I: the Kaplan-Meier method. *Nephron Clin Pract*. 2011;119(1):c83-c88. doi:10.1159/000324758
37. Harrington D. Linear rank tests in survival analysis. In: Armitage P, Colton T, eds. *Encyclopedia of Biostatistics*. 2nd ed. John Wiley & Sons, Ltd; 2005. Accessed June 17, 2024. <https://onlinelibrary.wiley.com/doi/abs/10.1002/0470011815.b2a11047>
38. Lundberg S, Lee SI. A unified approach to interpreting model predictions. *arXiv*. Preprint posted online May 22, 2017. Last revised November 25, 2017. doi:10.48550/arXiv.1705.07874
39. Sundararajan M, Taly A, Yan Q. Axiomatic attribution for deep networks. Proceedings of the 34th International Conference on Machine Learning, PMLR 70: 3319-3328; 2017. Sydney, NSW, Australia.
40. Hosmer DW, Lemeshow S. Goodness of fit tests for the multiple logistic regression model. *Commun Stat Theory Methods*. 1980;9(10):1043-1069. doi:10.1080/03610928008827941
41. Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Med Decis Making*. 2006;26(6):565-574. doi:10.1177/0272989X06295361
42. R: a Language and Environment for Statistical Computing. February 10, 2015. Accessed June 17, 2024. <https://www.gbif.org/tool/81287/r-a-language-and-environment-for-statistical-computing>
43. Ma B, Guo J, De Biase A, et al. PET/CT based transformer model for multi-outcome prediction in oropharyngeal cancer. *Radiother Oncol*. 2024;197:110368. doi:10.1016/j.radonc.2024.110368
44. Chen RJ, Lu MY, Wang J, et al. Pathomic fusion: an integrated framework for fusing histopathology and genomic features for cancer diagnosis and prognosis. *IEEE Trans Med Imaging*. 2022;41(4):757-770. doi:10.1109/TMI.2020.3021387
45. Sharma SJ, Wagner S, Reder HSF, et al. The 8th edition AJCC/UICC TNM staging for p16-positive oropharyngeal carcinoma: is there space for improvement? *Eur Arch Otorhinolaryngol*. 2018;275(12):3087-3091. doi:10.1007/s00405-018-5156-4
46. Cmelak AJ, Ferris RL, Chen AM, Seiwert T, Burtneis B. Treatment de-intensification for HPV-positive oropharynx cancer: what is currently acceptable? *J Clin Oncol*. 2021;39(24):2732-2733. doi:10.1200/JCO.21.00594
47. Chaturvedi AK, Engels EA, Pfeiffer RM, et al. Human papillomavirus and rising oropharyngeal cancer incidence in the United States. *J Clin Oncol*. 2011;29(32):4294-4301. doi:10.1200/JCO.2011.36.4596
48. Gillison ML, Chaturvedi AK, Anderson WF, Fakhry C. Epidemiology of human papillomavirus-positive head and neck squamous cell carcinoma. *J Clin Oncol*. 2015;33(29):3235-3242. doi:10.1200/JCO.2015.61.6995
49. Denis F, Garaud P, Bardet E, et al. Final results of the 94-01 French Head and Neck Oncology and Radiotherapy Group randomized trial comparing radiotherapy alone with concomitant radiochemotherapy in advanced-stage oropharynx carcinoma. *J Clin Oncol*. 2004;22(1):69-76. doi:10.1200/JCO.2004.08.021
50. Machtay M, Moughan J, Trotti A, et al. Factors associated with severe late toxicity after concurrent chemoradiation for locally advanced head and neck cancer: an RTOG analysis. *J Clin Oncol*. 2008;26(21):3582-3589. doi:10.1200/JCO.2007.14.8841
51. Ferris RL, Flamand Y, Weinstein GS, et al. Phase II randomized trial of transoral surgery and low-dose intensity modulated radiation therapy in resectable p16+ locally advanced oropharynx cancer: an ECOG-ACRIN Cancer Research Group Trial (E3311). *J Clin Oncol*. 2022;40(2):138-149. doi:10.1200/JCO.21.01752
52. Lim YX, Mierzwa ML, Sartor MA, D'Silva NJ. Clinical, morphologic and molecular heterogeneity of HPV-associated oropharyngeal cancer. *Oncogene*. 2023;42(40):2939-2955. doi:10.1038/s41388-023-02819-y

53. Zhang N, Liang R, Gensheimer MF, et al. Early response evaluation using primary tumor and nodal imaging features to predict progression-free survival of locally advanced non-small cell lung cancer. *Theranostics*. 2020;10(25):11707-11718. doi:10.7150/thno.50565
54. Haider SP, Sharaf K, Zeevi T, et al. Prediction of post-radiotherapy locoregional progression in HPV-associated oropharyngeal squamous cell carcinoma using machine-learning analysis of baseline PET/CT radiomics. *Transl Oncol*. 2021;14(1):100906. doi:10.1016/j.tranon.2020.100906
55. Boehm KM, Aherne EA, Ellenson L, et al; MSK MIND Consortium. Multimodal data integration using machine learning improves risk stratification of high-grade serous ovarian cancer. *Nat Cancer*. 2022;3(6):723-733. doi:10.1038/s43018-022-00388-9
56. Zeng PYF, Cecchini MJ, Barrett JW, et al. Immune-based classification of HPV-associated oropharyngeal cancer with implications for biomarker-driven treatment de-intensification. *EBioMedicine*. 2022;86:104373. doi:10.1016/j.ebiom.2022.104373
57. Wilde DC, Castro PD, Bera K, et al. Oropharyngeal cancer outcomes correlate with p16 status, multinucleation and immune infiltration. *Mod Pathol*. 2022;35(8):1045-1054. doi:10.1038/s41379-022-01024-8
58. Liu X, Liu P, Chernock RD, et al. A prognostic gene expression signature for oropharyngeal squamous cell carcinoma. *EBioMedicine*. 2020;61:102805. doi:10.1016/j.ebiom.2020.102805
59. Liu X, Liu P, Chernock RD, et al. A MicroRNA expression signature as prognostic marker for oropharyngeal squamous cell carcinoma. *J Natl Cancer Inst*. 2021;113(6):752-759. doi:10.1093/jnci/djaa161

SUPPLEMENT 1.

eTable 1. Treatment Details for Patients Included in this Study

eTable 2. CT Imaging Parameters for the Three Cohorts

eTable 3. Clinical Characteristics Within the High- and Low-SwinScore Groups for Patients in D₁+D₂+D₃

eTable 4. Point-Biserial Correlation Coefficients Computed by Non-parametric Wilcoxon Rank Sum Test Between Each Individual Imaging Biomarker (ie, Swin-Score, Swin-PT, Swin-LN) and the Clinicopathologic Factors (Columns) on D₃

eTable 5. Comparative Analysis of Prognostic Performance (Measured by C-Index on D₃ Using DFS as Endpoint) on Models Employing Different Feature Extraction Backbones, Regions of Interest, and Embedding Fusion Strategies

eTable 6. Clinical Characteristics According to Treatment Groups for All the Patients in This Study

eFigure 1. Patient Selection and Allocation Diagram for Patients in D₃

eFigure 2. Workflow of This Study

eFigure 3. Loss and C-Index Changes According to Epoch

eFigure 4. Barplot Visualizing Risk Scores on D₃

eFigure 5. The Kaplan-Meier Survival Analysis of SwinScore for p16+ OPSCC Stratification on D₁ Using DFS (A), OS (B) and LRF (C) as Endpoints and on D₂ Using DFS (D), OS (E) and LRF (F) as Endpoints

eFigure 6. The Kaplan-Meier Survival Analysis of SwinScore for AJCC Stage I Patients Using DFS (A, B), OS (C, D) and LRF (E, F) as Endpoints

eFigure 7. The Kaplan-Meier Survival Analysis of SwinScore for AJCC Stage II Patients Using DFS (A, B), OS (C, D) and LRF (E, F) as Endpoints

eFigure 8. The Kaplan-Meier Survival Analysis of SwinScore for AJCC Stage III Patients Using DFS (A, B), OS (C, D) and LRF (E, F) as Endpoints

eFigure 9. Calibration Curves to Compare SwinScore Predicted DFS (A) and OS (B) Probabilities and the Actual Survival Rates on D₃

eFigure 10. (A) Mean SHAP Values Were Converted to Proportion for Each Variable in Multivariate Cox Analysis Using DFS as Outcome. (B) Mean SHAP Values Across Different Outcome Endpoints

eFigure 11. Boxplot Visualizing Risk Score Differences Between Patients Who Had Event for 3-Year DFS and Those Who are Censored on D₃ (A). DFS Death/Failure Rate Within Each Dichotomized Risk Group (B)

eFigure 12. Primary Tumor and Metastatic Cervical Lymph Node Carry Complementary Prognostic Values

eFigure 13. The Correlation Analysis of SwinScore's Penultimate Layer Features Revealed Several Key Pathways Significantly Associated With the Gene Sets, Which Were Visualized in a Heatmap

eFigure 14. The Kaplan-Meier Survival Analysis for Comparing DFS (A), OS (B) and LRF (C) Between the Two Treatment Arms Among all the 811 Patients Used in this Study

eFigure 15. SwinScore as a Predictive Biomarker of Added Benefit Of Chemotherapy

eMethods 1. CT Imaging Acquisition and Manual Annotations

eMethods 2. CT Image Preprocessing

eMethods 3. Hierarchical Embedding With Swin Transformer

eMethods 4. Attention-Based Fusion and Training Details

eMethods 5. Outcome Definitions

eMethods 6.
eReferences.

[SUPPLEMENT 2.](#)
Data Sharing Statement